OSSP (Operating Systems And Systems Programming) - 25CS2104E

1. Develop a C program that demonstrates how a Linux operating system executes a command entered by a user that 1. Accept a Linux command as input. 2. Create a child process using fork(). 3. Execute the command in the child process using an appropriate exec() system call. 4. Allow the parent process to wait for the child using wait (). 5. Display the Process ID (PID) of both parent and child processes.

   Using Linux terminal commands (uname, lscpu, lsblk, ps, top), investigate the relationship between hardware resources and operating system services. Prepare a report explaining how the OS abstracts CPU, memory, storage, and I/O devices

```c
#include <stdio.h>
#include <stdlib.h>
#include <unistd.h>
#include <sys/wait.h>

int main()
{
    char cmd[100];

    printf("Enter Linux command: ");
    scanf("%s", cmd);
    pid_t pid = fork();
    if(pid == 0)
    {
        printf("Child PID : %d\n", getpid());
        execlp(cmd, cmd, NULL);
        perror("Execution Failed");
    }
    else if(pid > 0)
    {
        printf("Parent PID : %d\n", getpid());
        wait(NULL);
        printf("Child process completed.\n");
    }
    else
    {
        printf("Fork Failed\n");
    }
    return 0;
}
```

```
hemasai@Hemasai: ~

hemasai@Hemasai:~$ nano task1.c
hemasai@Hemasai:~$ gcc task1.c -otask1
hemasai@Hemasai:~$ ./task1
Enter Linux command: uname -a
Parent PID : 652
Child PID : 653
Linux
Child process completed.
hemasai@Hemasai:~$
```

```
Child process completed.
hemasai@Hemasai:~$ lscpu
Architecture:                         x86_64
  CPU op-mode(s):                     32-bit, 64-bit
  Address sizes:                      39 bits physical, 48 bits virtual
  Byte Order:                         Little Endian
CPU(s):                               12
  On-line CPU(s) list:                0-11
Vendor ID:                            GenuineIntel
  Model name:                         12th Gen Intel(R) Core(TM) i5-1235U
    CPU family:                       6
    Model:                            154
    Thread(s) per core:               2
    Core(s) per socket:               6
    Socket(s):                        1
    Stepping:                         4
    BogoMIPS:                         4992.00
    Flags:                            fpu vme de pse tsc msr pae mce cx8
                                      apic sep mtrr pge mca cmov pat pse3
                                      6 clflush mmx fxsr sse sse2 ss ht s
                                      yscall nx pdpe1gb rdtscp lm constan
                                      t_tsc rep_good nopl xtopology tsc_r
                                      eliable nonstop_tsc cpuid tsc_known
                                      _freq pni pclmulqdq vmx ssse3 fma c
                                      x16 pcid sse4_1 sse4_2 x2apic movbe
                                       popcnt tsc_deadline_timer aes xsav
                                      e avx f16c rdrand hypervisor lahf_l
                                      m abm 3dnowprefetch ssbd ibrs ibpb
                                      stibp ibrs_enhanced tpr_shadow ept
                                      vpid ept_ad fsgsbase tsc_adjust bmi
                                      1 avx2 smep bmi2 erms invpcid rdsee
                                      d adx smap clflushopt clwb sha_ni x
                                      saveopt xsavec xgetbv1 xsaves avx_v
                                      nni vnmi umip waitpkg gfni vaes vpc
                                      lmulqdq rdpid movdiri movdir64b fsr
                                      m md_clear serialize ibt flush_l1d
                                      arch_capabilities
Virtualization features:
  Virtualization:                     VT-x
  Hypervisor vendor:                  Microsoft
  Virtualization type:                full
Caches (sum of all):
  L1d:                                288 KiB (6 instances)
  L1i:                                192 KiB (6 instances)
  L2:                                 7.5 MiB (6 instances)
  L3:                                 12 MiB (1 instance)
NUMA:
  NUMA node(s):                       1
  NUMA node0 CPU(s):                  0-11
Vulnerabilities:
  Gather data sampling:               Not affected
  Ghostwrite:                         Not affected
  Indirect target selection:          Not affected
  Itlb multihit:                      Not affected
  L1tf:                               Not affected
  Mds:                                Not affected
  Meltdown:                           Not affected
  Mmio stale data:                    Not affected
  Old microcode:                      Not affected
  Reg file data sampling:             Mitigation; Clear Register File
  Retbleed:                           Mitigation; Enhanced IBRS
  Spec rstack overflow:               Not affected
  Spec store bypass:                  Mitigation; Speculative Store Bypas
                                      s disabled via prctl
  Spectre v1:                         Mitigation; usercopy/swapgs barrier
                                      s and __user pointer sanitization
  Spectre v2:                         Mitigation; Enhanced / Automatic IB
                                      RS; IBPB conditional; PBRSB-eIBRS S
                                      W sequence; BHI BHI_DIS_S
  Srbds:                              Not affected
  Tsa:                                Not affected
  Tsx async abort:                    Not affected
  Vmscape:                            Not affected
hemasai@Hemasai:~$
```

```
> ^C
hemasai@Hemasai:~$ lsblk
NAME MAJ:MIN RM   SIZE RO TYPE MOUNTPOINTS
sda    8:0    0 356.9M  1 disk
sdb    8:16   0 159.5M  1 disk
sdc    8:32   0     2G  0 disk [SWAP]
sdd    8:48   0     1T  0 disk /mnt/wslg/distro
                                /
hemasai@Hemasai:~$
```

```
hemasai@Hemasai:~$ ps -e
    PID TTY          TIME CMD
      1 ?        00:00:01 systemd
      2 hvc0     00:00:00 init-systemd(Ub
      6 hvc0     00:00:00 init
     49 ?        00:00:00 systemd-journal
     81 ?        00:00:00 systemd-resolve
     90 ?        00:00:00 systemd-udevd
    150 ?        00:00:00 chronyd-starter
    151 ?        00:00:00 cron
    152 ?        00:00:00 dbus-daemon
    156 ?        00:00:00 networkd-dispat
    164 ?        00:00:00 systemd-logind
    198 tty1     00:00:00 agetty
    199 ?        00:00:00 rsyslogd
    228 ?        00:00:00 chronyd
    245 ?        00:00:00 chronyd
    255 ?        00:00:00 unattended-upgr
    344 ?        00:00:00 login
    386 ?        00:00:00 systemd
    388 ?        00:00:00 (sd-pam)
    419 pts/1    00:00:00 bash
    588 ?        00:00:00 SessionLeader
    590 ?        00:00:00 Relay(593)
    593 pts/0    00:00:00 bash
    711 pts/0    00:00:00 ps
hemasai@Hemasai:~$
```

```
    711 pts/0    00:00:00 ps
hemasai@Hemasai:~$ top
top - 09:26:06 up 12 min,  1 user,  load average: 0.03, 0.06, 0.01
Tasks:  24 total,   1 running,  23 sleeping,   0 stopped,   0 zombie
%Cpu(s):  0.0 us,  0.0 sy,  0.0 ni, 99.9 id,  0.1 wa,  0.0 hi,  0.0 si,  0.0 st
MiB Mem :   7776.6 total,   7273.3 free,    524.4 used,    123.9 buff/cache
MiB Swap:   2048.0 total,   2048.0 free,      0.0 used.   7252.2 avail Mem

    PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM     TIME+ COMMAND
      1 root      20   0   24212  15344  11564 S   0.0   0.2   0:01.17 systemd
      2 root      20   0    3180   2208   2072 S   0.0   0.0   0:00.03 init-syst+
      6 root      20   0    3180   2044   1940 S   0.0   0.0   0:00.00 init
     49 root      19  -1   50400  16816  15528 S   0.0   0.2   0:00.33 systemd-j+
     81 systemd+  20   0   22540  14600  11980 S   0.0   0.2   0:00.12 systemd-r+
     90 root      20   0   35360  12312   9228 S   0.0   0.2   0:00.50 systemd-u+
    150 root      20   0    2888   1948   1820 S   0.0   0.0   0:00.08 chronyd-s+
    151 root      20   0    4472   3140   2896 S   0.0   0.0   0:00.01 cron
    152 message+  20   0    8824   5384   4700 S   0.0   0.1   0:00.13 dbus-daem+
    156 root      20   0   44092  29692  14552 S   0.0   0.4   0:00.55 networkd-+
    164 root      20   0   18432   9440   8196 S   0.0   0.1   0:00.25 systemd-l+
    198 root      20   0    5492   2768   2580 S   0.0   0.0   0:00.01 agetty
    199 syslog    20   0  220548   5936   4656 S   0.0   0.1   0:00.16 rsyslogd
    228 _chrony   20   0   20516   7852   7000 S   0.0   0.1   0:00.10 chronyd
    245 _chrony   20   0   12092   2404   1760 S   0.0   0.0   0:00.00 chronyd
    255 root      20   0  123456  32648  18044 S   0.0   0.4   0:00.19 unattende+
    344 root      20   0    8648   5628   4784 S   0.0   0.1   0:00.02 login
    386 hemasai   20   0   22336  13588  10968 S   0.0   0.2   0:00.14 systemd
    388 hemasai   20   0   22908   4088   2088 S   0.0   0.1   0:00.00 (sd-pam)
    419 hemasai   20   0    6136   5476   3928 S   0.0   0.1   0:00.02 bash
    588 root      20   0    3196   1124    980 S   0.0   0.0   0:00.00 SessionLe+
    590 root      20   0    3196   1256   1108 S   0.0   0.0   0:00.03 Relay(593)
    593 hemasai   20   0    6268   5524   3848 S   0.0   0.1   0:00.11 bash
    723 hemasai   20   0    8184   6016   3900 R   0.0   0.1   0:00.01 top
```

2. Develop a C program that uses the system calls open(), read(), write(), and close() to copy the contents of one file to another. Explain how control transitions between user space and kernel space during execution.

   Use the strace utility to trace all system calls generated by the command cat sample.txt. Analyze the sequence of system calls and identify the kernel services involved.